CataTriage AI Two Page Summary

CodeJediX Implementation milestone 30 October 2026

CataTriage AI is a working research prototype for reviewing cataract related information and coordinating a doctor approved hospital workflow. It combines two real data research models, protected doctor admin and technician portals, four auditable processing stages and a native doctor Android app. Its measured model performance is limited, so it does not make clinical decisions.

What a user can do

User	Implemented functions
Doctor	Review assigned patients and per eye original images, record symptoms and notes, inspect results, enter a tier, approve a month, message, manage profile and password.
Hospital admin	Manage accounts and patients, inspect approved queue and capacity, choose a free date inside the approved month, resolve leave and view audit history.
Technician	Register structured information and upload left right or unknown side images with immediate technical quality feedback.

How the system works

Preprocessing checks input quality and completeness. Inference runs the model for the declared modality. Safety checks flag uncertainty and unsupported conditions. Reporting presents the result and audit trail for review. These are deterministic software stages, not independent medical agents.

Scheduling is doctor tier and month approval followed by admin date confirmation. Model confidence does not determine urgency or a safe waiting period. Changed evidence requires fresh month review. The floating red bell persists while approved months lack exact dates. A built in navigation guide supplies role specific links, workflow help, model metrics and capture checklists.

Stack and delivery

FastAPI and Jinja2 provide one Python web and mobile API codebase. PyTorch and exported NumPy tree inference run on CPU or CUDA. Three SQLite files support local use; cloud mode uses three private Supabase schemas. Vercel hosts the app. Optional Gemma explanations have a configured free fallback. The doctor app is native Java Android 8 or newer, with server side inference.

Source code, model checkpoints, SQL, tests, training utilities, APK and setup instructions accompany this summary. The local demo starts without raw datasets, cloud keys or production passwords after dependencies are installed. Every supplied PDF has an editable Word counterpart.

Results data and judging guide

Model	Training	Validation	Test	Previous test
Grading v2	70.79%	54.36%	56.04%	53.69%
Fusion v2	72.14%	65.00%	55.00%	52.50%

Grading gained seven exact predictions on 298 test participants; fusion gained one on 40 images. Both also improved test macro F1. The intervals are wide and overlap prior scores: grading about 50.7 to 61.7 percent, fusion 37.5 to 71.4 percent. Normal recall is still only 2 of 8. A trial with 99 percent training accuracy was rejected for validation overfitting.

The data and its age

Grading uses 1,986 Taizhou participants from a study conducted in July to September 2012 and published through Dryad in 2016. The split is 1,390 training, 298 validation and 298 test. It is one retrospective cohort with a derived maximum bilateral grade, not multi cohort clinical validation.

Fusion uses the 200 image CatScreen public clean sample downloaded on 26 October 2026, split 140 training, 20 validation and 40 test with disjoint participant IDs. Collection dates were not verified from the sample. Its context is acquisition metadata, not full clinical history. The two cohorts are not linked; no chronological, external or prospective validation is established.

Honest comparison with the proposal

The interface, human approval workflow, role stores, reports, messaging, profile features, alerts, chatbot and doctor APK are implemented. Verified image linked severity, full clinical fusion, a trained urgency or safe wait model and clinical deployment readiness remain unproven. Automatic WhatsApp needs a verified Business configuration; the navigation guide is not a medical LLM. Synthetic workflow fixtures are distinct from real model evidence.

How judges can inspect it

Read the implementation report, run the README local setup and tools/judge_smoke.py, and try doctor month approval followed by admin date selection. Inspect the current model evidence and four agent trace, then try “open calendar” in the guide. The recorded workspace run passed 110 backend tests; a source only checkout skips one optional raw image test. Browser checks and live non persistent inference also passed.

Website https://catatriage-ai.vercel.app
Repository https://github.com/CodeJediX/CataTriage-AI
Data sources https://doi.org/10.5061/dryad.gt410 and the official CatScreen dataset page.
The local fixture accounts are in README. Hosted account credentials are not distributed.